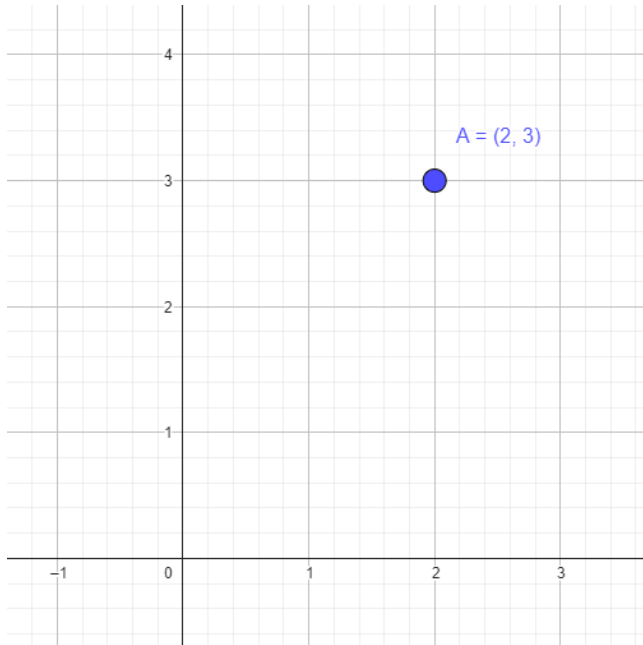


Points:

$A(x,y)$

$A(2,3)$



Distance between two points:

$A(2,3)$

$B(5,7)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(5 - 2)^2 + (7 - 3)^2}$$

$$d = \sqrt{3^2 + 4^2}$$

$$d = \sqrt{9 + 16}$$

$$d = \sqrt{25}$$

$$d = 5$$

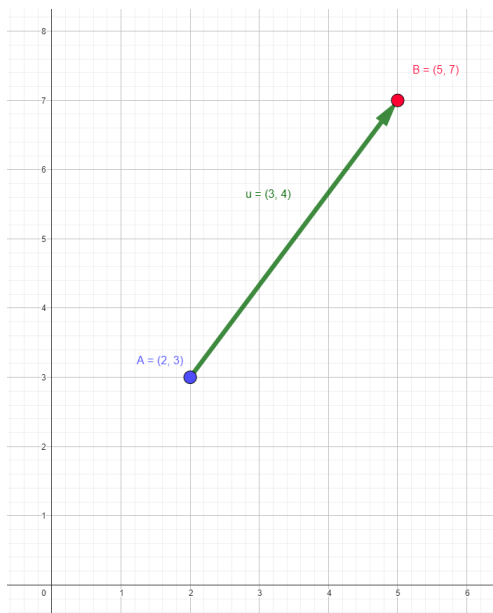
	$A = (2, 3)$
	$B = (5, 7)$
	$d = \text{Distance}(A, B)$
	$= 5$

Vector pointing from **A** to **B**:

$$\vec{u} = (B_x - A_x, B_y - A_y)$$

$$\vec{u} = (5 - 2, 7 - 3)$$

$$\vec{u} = (3, 4)$$



$$\mathbf{u} = \text{Vector}(\mathbf{A}, \mathbf{B})$$

	$A = (2, 3)$
	$B = (5, 7)$
	$\mathbf{u} = \text{Vector}(A, B)$
	$= \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

Adding two vectors:

$$u = (3, 4)$$

$$v = (5, 2)$$

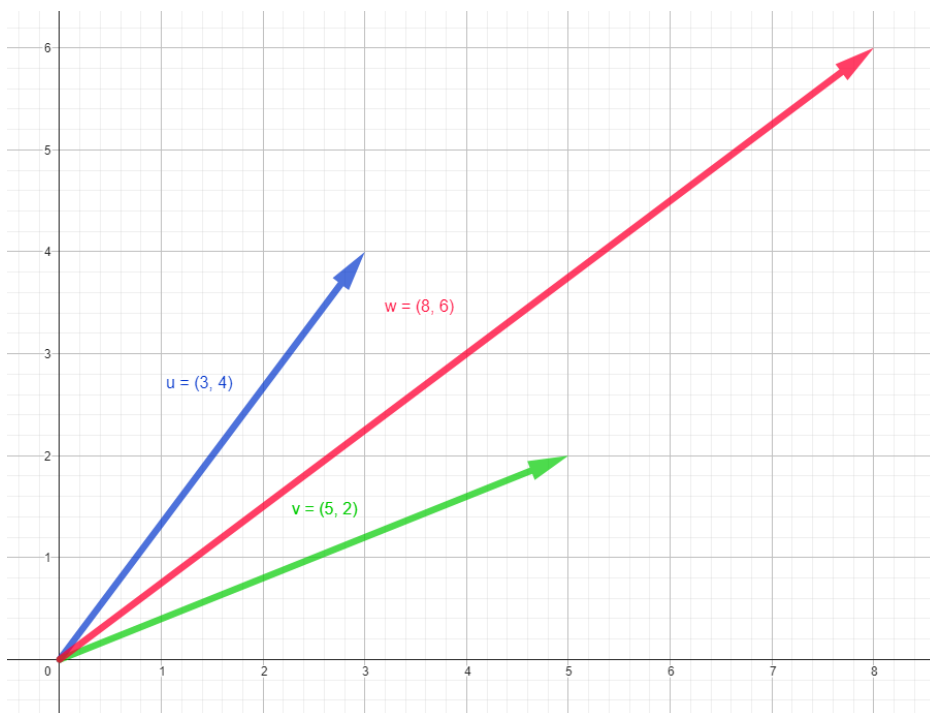
$$\vec{u} + \vec{v} = (u_x + v_x, u_y + v_y)$$

$$\vec{u} + \vec{v} = (3 + 5, 4 + 2)$$

$$\vec{u} + \vec{v} = (8, 6)$$

$$w = u + v$$

●	$u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$
●	$v = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$
●	$w = u + v$ $= \begin{pmatrix} 8 \\ 6 \end{pmatrix}$



Length of a vector:

$$||\vec{u}|| = \sqrt{(u_x)^2 + (u_y)^2}$$

$$||\vec{u}|| = \sqrt{3^2 + 4^2}$$

$$||\vec{u}|| = \sqrt{9 + 16}$$

$$||\vec{u}|| = \sqrt{25}$$

$$||\vec{u}|| = 5$$

$$a = \text{Length}(\mathbf{u})$$

Multiplying a vector by a scalar:

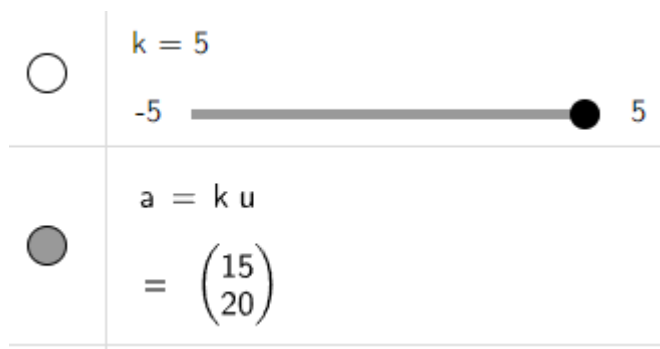
$$k = 5$$

$$k \cdot \vec{u} = (k \cdot u_x, k \cdot u_y)$$

$$k \cdot \vec{u} = (5 \cdot 3, 5 \cdot 4)$$

$$k \cdot \vec{u} = (15, 20)$$

$$\mathbf{a} = k \cdot \mathbf{u}$$



Inner product of two vectors:

$$\vec{u} \cdot \vec{v} = (u_x \cdot v_x + u_y \cdot v_y)$$

$$\vec{u} \cdot \vec{v} = (3 \cdot 5 + 4 \cdot 2)$$

$$\vec{u} \cdot \vec{v} = (15 + 8)$$

$$\vec{u} \cdot \vec{v} = 23$$

$$a = u^*v$$

$$a = \text{Dot}(u,v)$$

	$u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$
	$v = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$
	$a = u \cdot v$
	$= 23$

Angle between two vectors:

$$\cos(\alpha) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \cdot \|\vec{v}\|}$$

$$\cos(\alpha) = \frac{23}{5 \cdot 5.39}$$

$$\cos(\alpha) = 0.85$$

$$\cos^{-1}(0.85) = 31.79^\circ$$

$$a = \text{Dot}(u,v)/\text{Length}(u)*\text{Length}(v)$$

$$\text{acos}(a)$$

	$u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$
	$v = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$
	$a = \frac{u \cdot v}{\text{Length}(u) \text{Length}(v)}$ $= 0.85$
	$\alpha = \cos^{-1}(0.85)$ $= 31.79^\circ$

Triple product of vectors:

$$\vec{a} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix}, \vec{b} = \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix}, \vec{c} = \begin{pmatrix} c_x \\ c_y \\ c_z \end{pmatrix}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c})$$

$$\begin{pmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{pmatrix}$$

$$\vec{b} \times \vec{c} = \begin{pmatrix} b_y c_z - b_z c_y \\ b_z c_x - b_x c_z \\ b_x c_y - b_y c_x \end{pmatrix}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = a_x(b_y c_z - b_z c_y) + a_y(b_z c_x - b_x c_z) + a_z(b_x c_y - b_y c_x)$$

$$\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \vec{c} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$\vec{b} \times \vec{c} = \begin{pmatrix} 5 \cdot 9 - 6 \cdot 8 \\ 6 \cdot 7 - 4 \cdot 9 \\ 4 \cdot 8 - 5 \cdot 7 \end{pmatrix}$$

$$\vec{b} \times \vec{c} = \begin{pmatrix} 45 - 48 \\ 42 - 36 \\ 32 - 35 \end{pmatrix}$$

$$\vec{b} \times \vec{c} = \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = 1 \cdot (-3) + 2 \cdot 6 + 3 \cdot (-3)$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = -3 + 12 - 9$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

$$\mathbf{a} = (1, 2, 3)$$

$$\mathbf{b} = (4, 5, 6)$$

$$\mathbf{c} = (7, 8, 9)$$

$$\mathbf{u} = \text{cross}(\mathbf{b}, \mathbf{c})$$

$$\mathbf{d} = \mathbf{a} * \mathbf{u}$$

●	$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$
●	$\mathbf{b} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$
●	$\mathbf{c} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$
●	$\mathbf{u} = \mathbf{b} \otimes \mathbf{c}$ $= \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$
	$\mathbf{d} = \mathbf{a} * \mathbf{u}$ $= 0$